

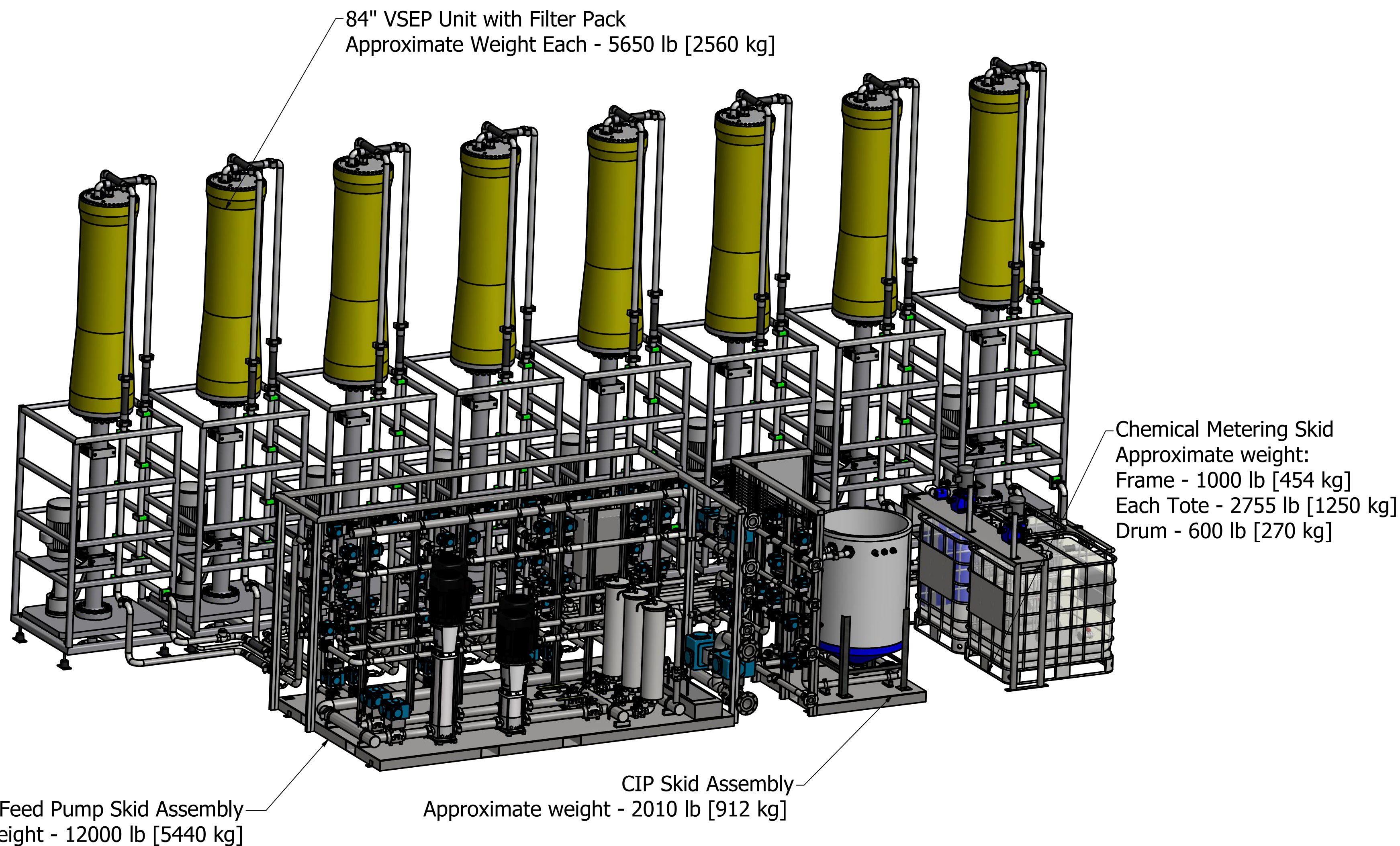
4

3

2

1

ISO VIEW

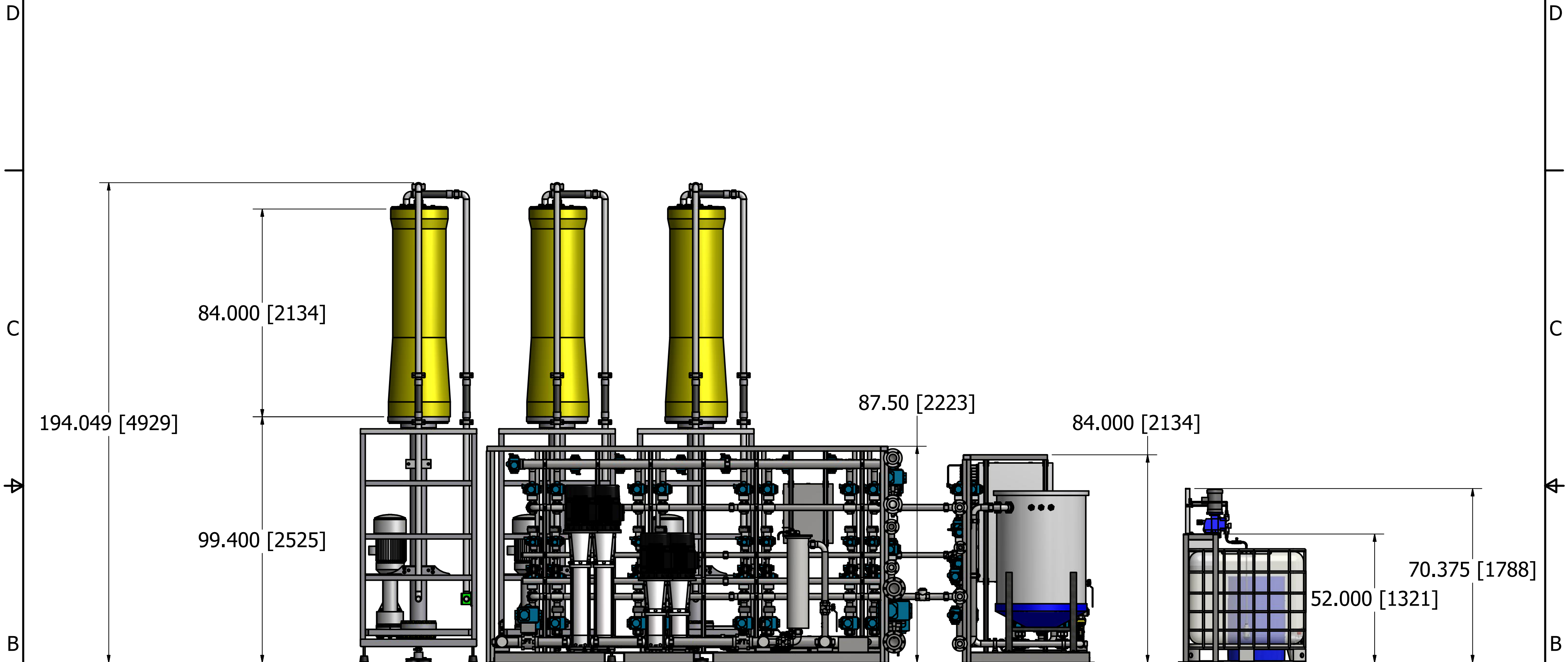


84" 8 Module System

NOTES:

1. New Logic Research confidential material.
2. Slight differences between drawings and actual system may be attributed to New Logic Research Continuously evolving and improving its technology.
3. All dimensions are shown in inches [mm] and for reference only.

FRONT VIEW
VSEP 6-8 (VSEP 1-5 not shown for clarity)



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4

3

2

1

TOP VIEW

D

D

C

C

B

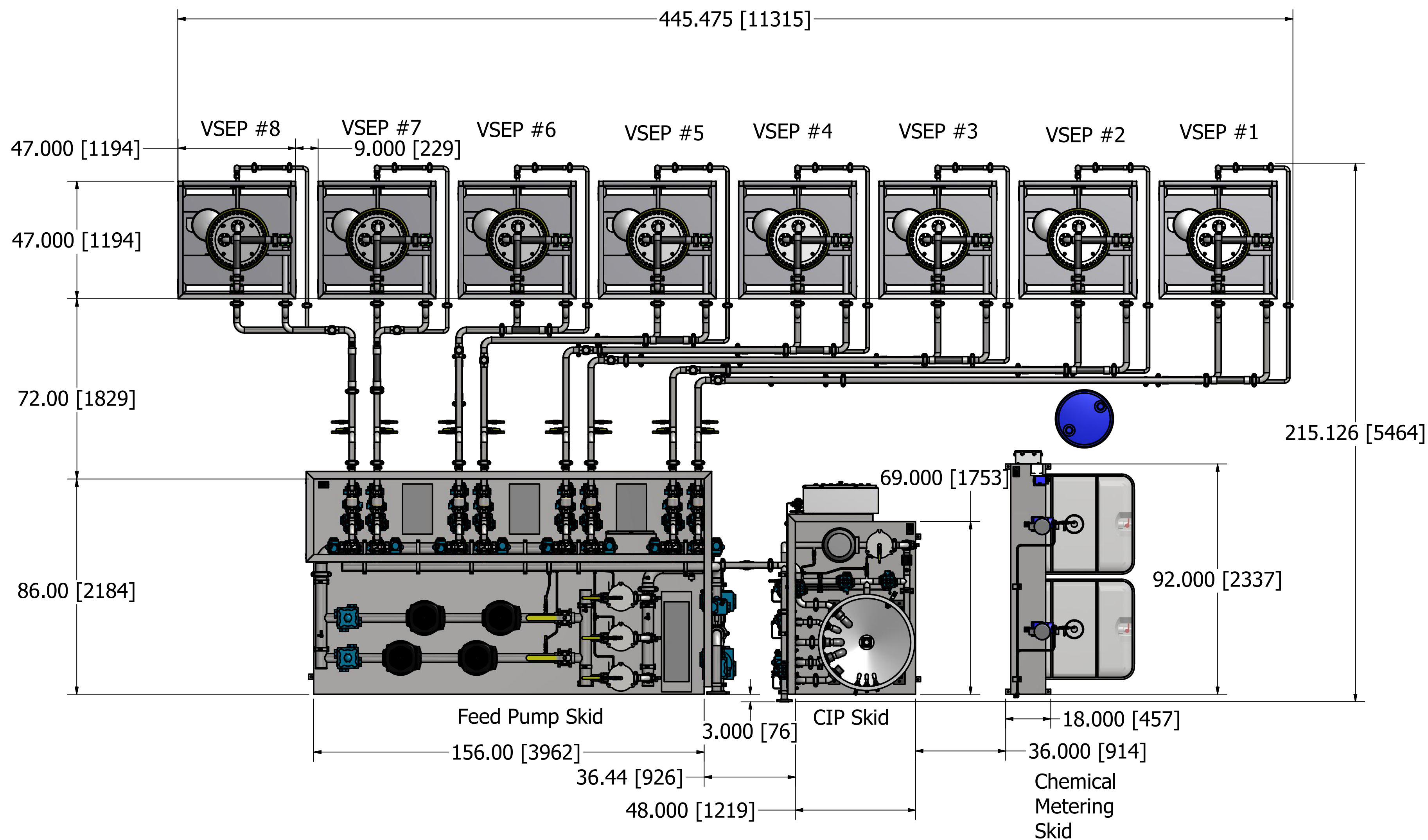
B

B

B

A

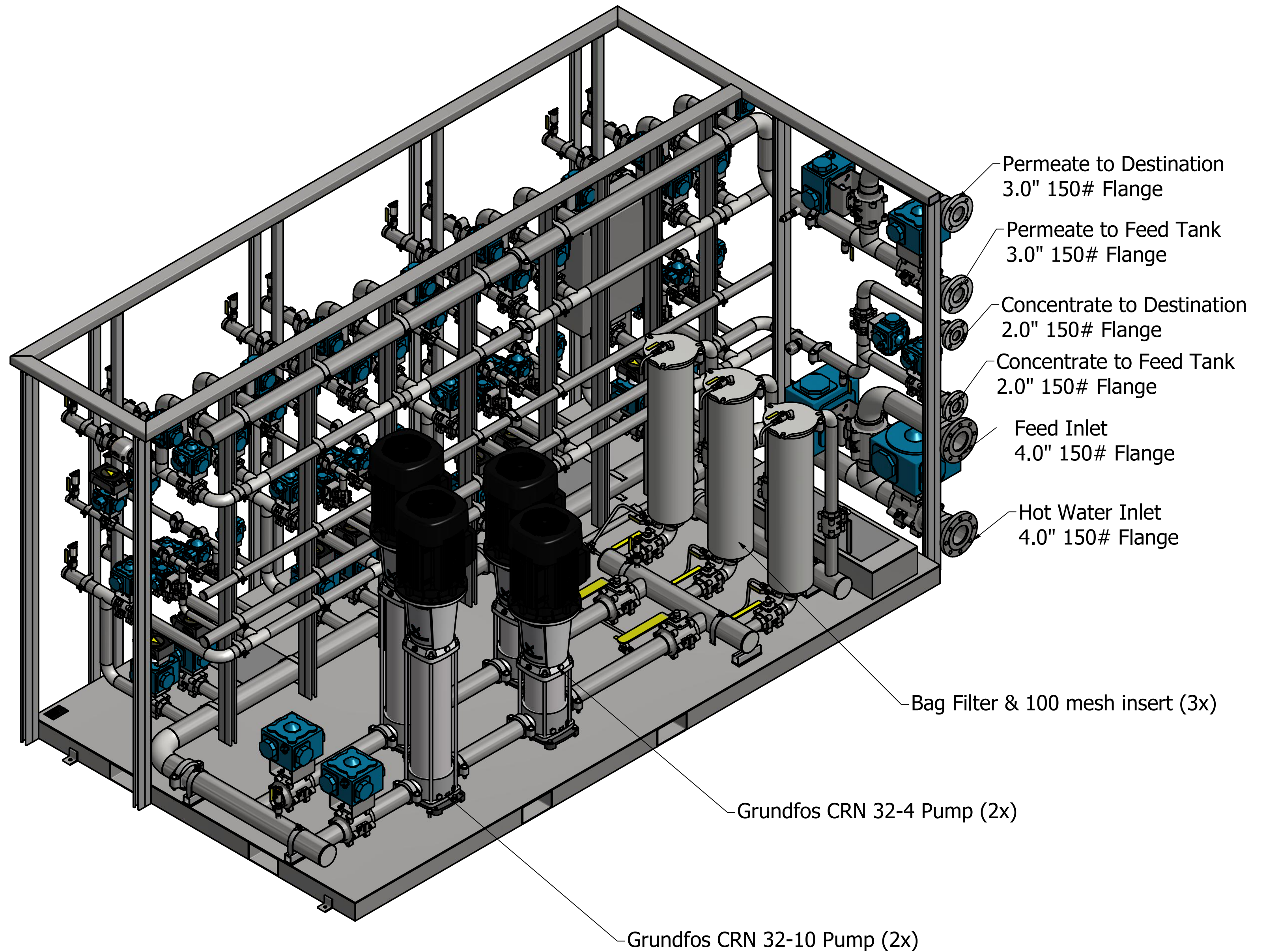
A



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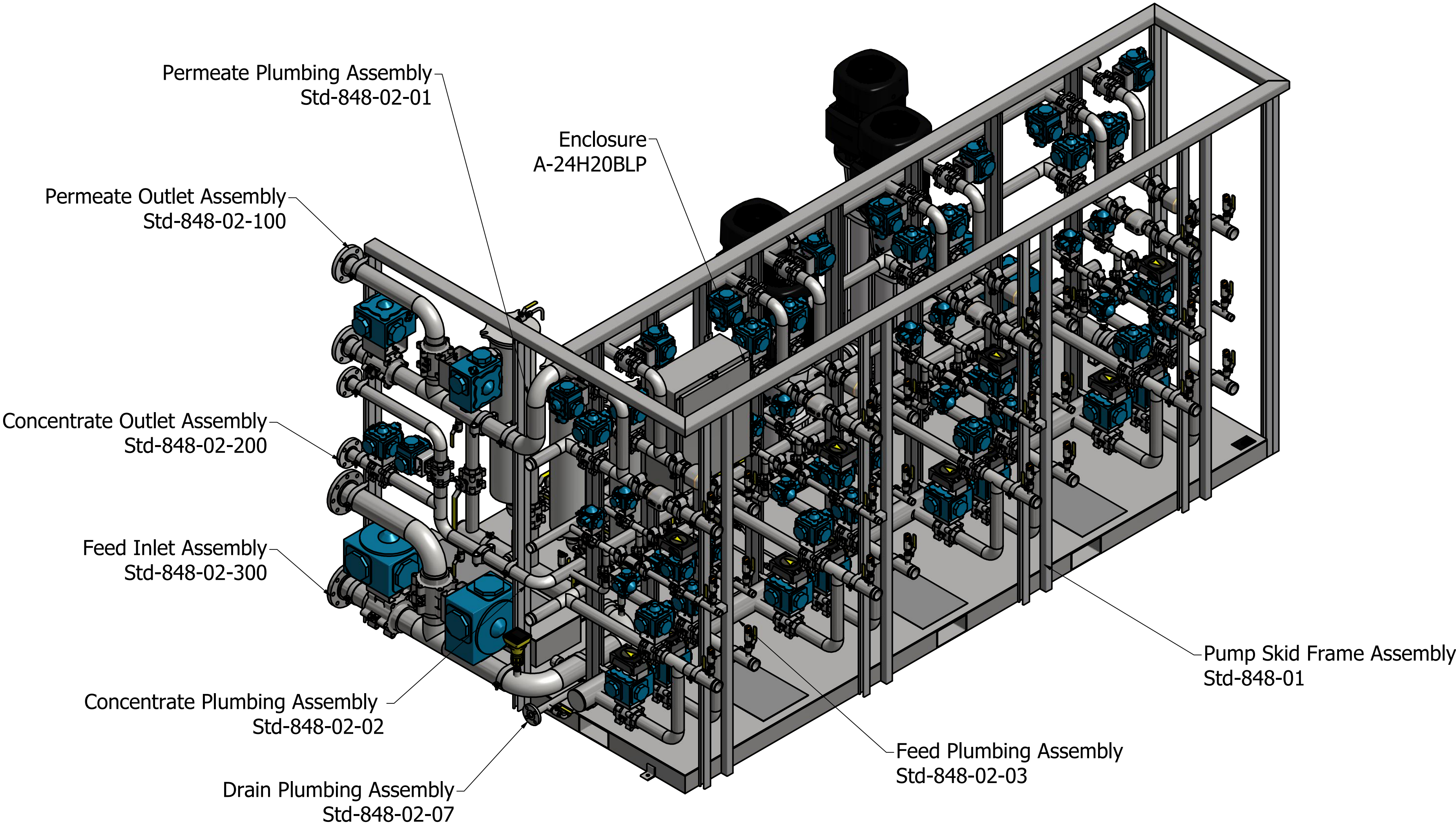
FRONT ISO VIEW



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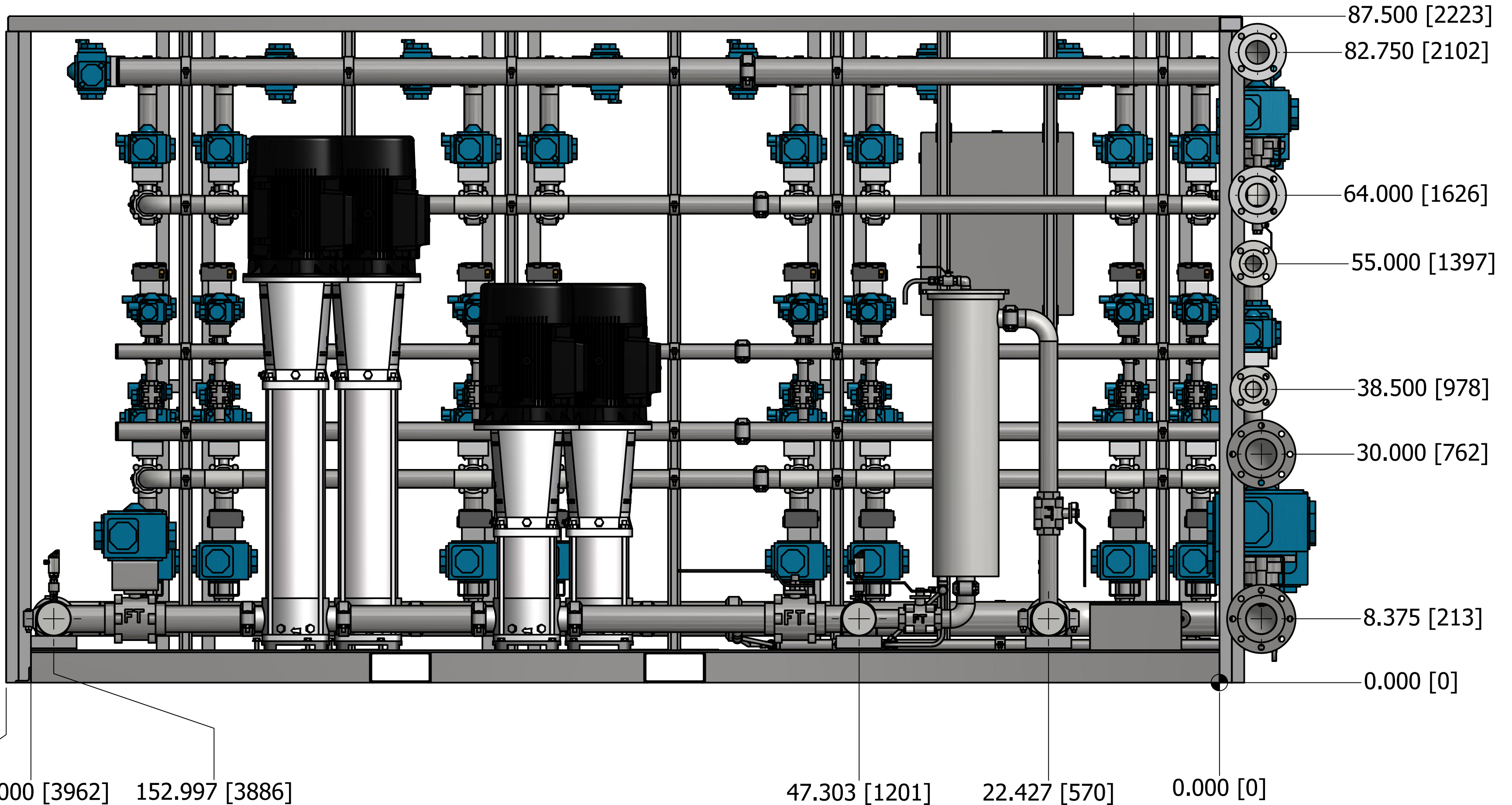
BACK ISO VIEW



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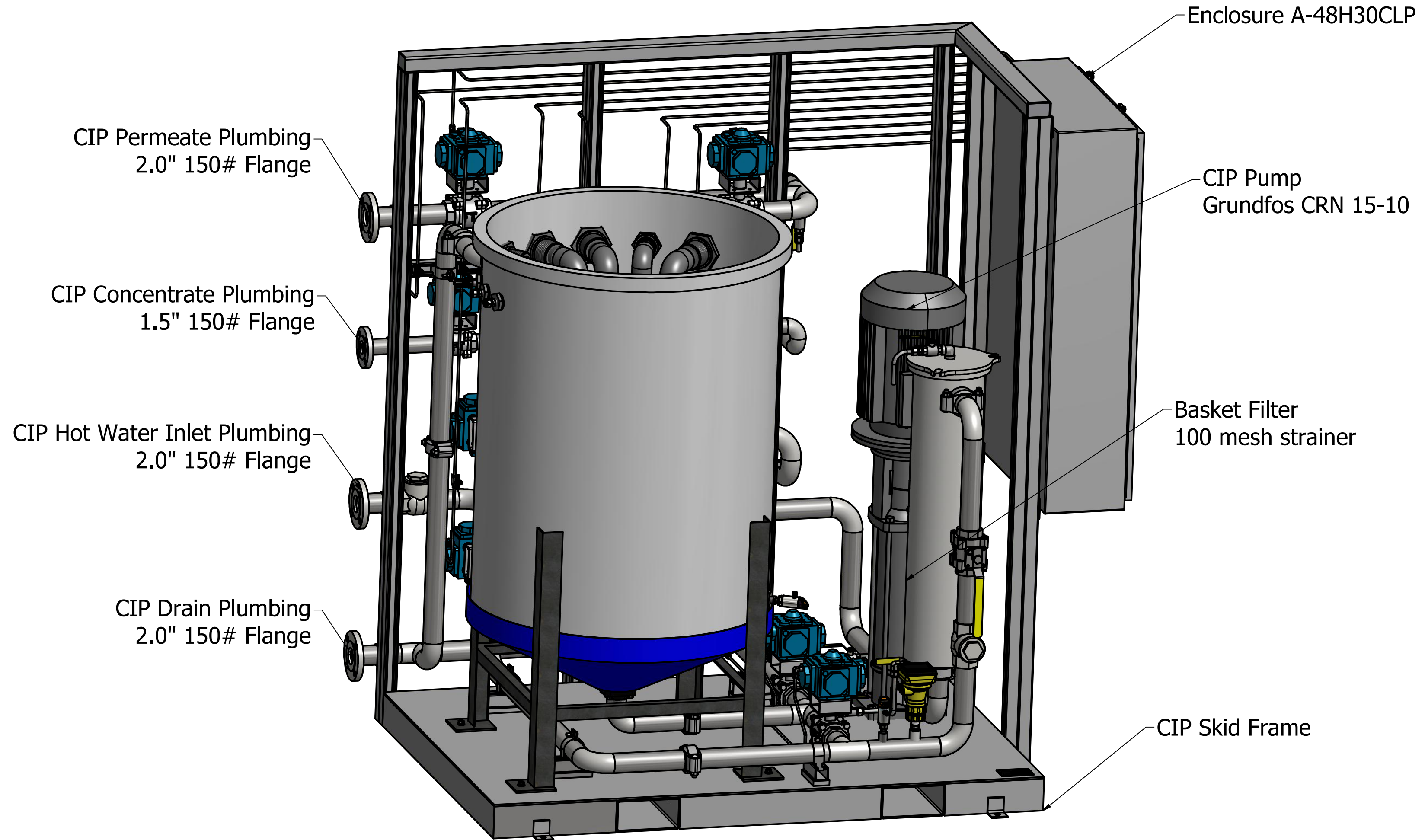
FRONT VIEW



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FRONT ISO VIEW

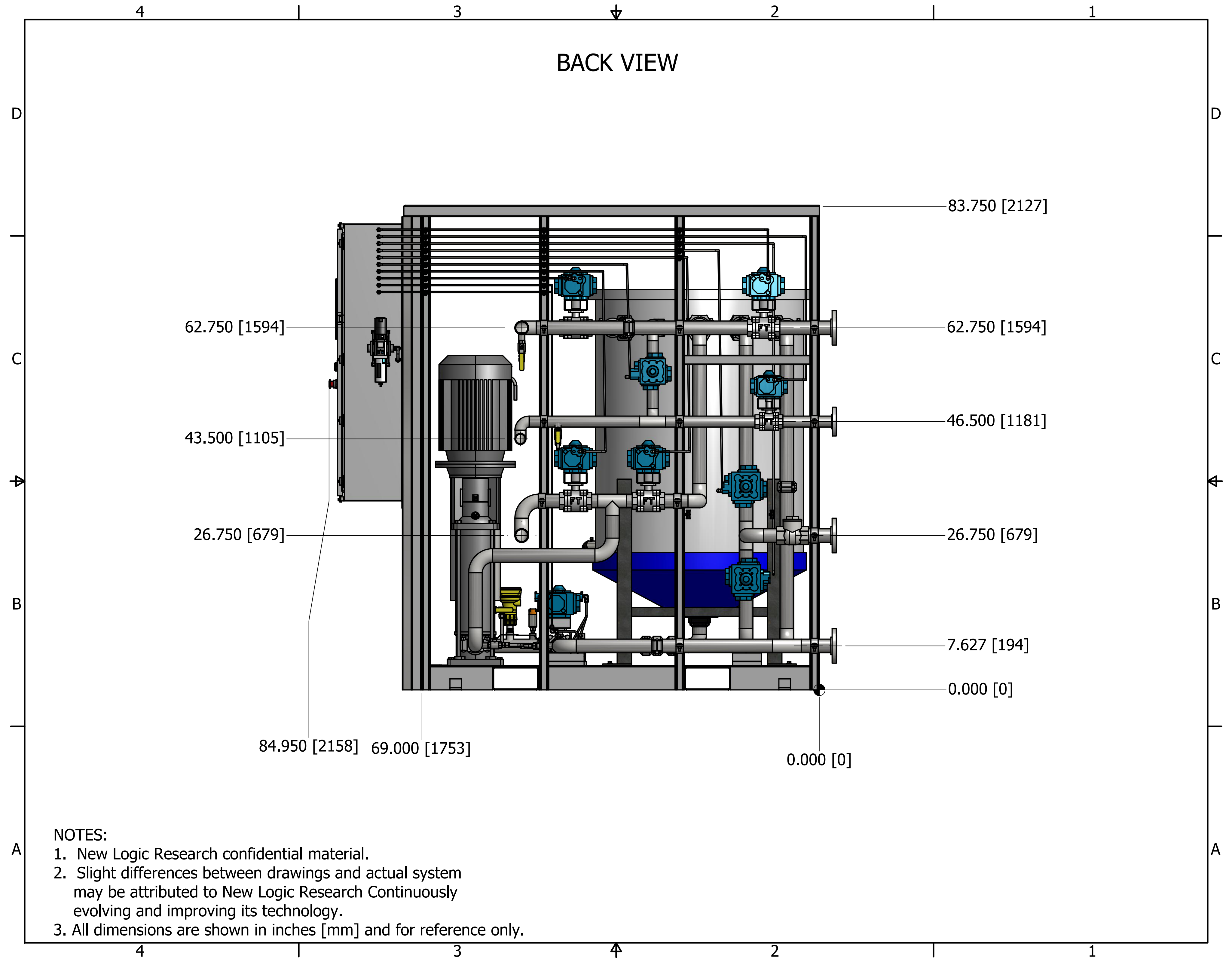


Standard CIP Skid Assembly

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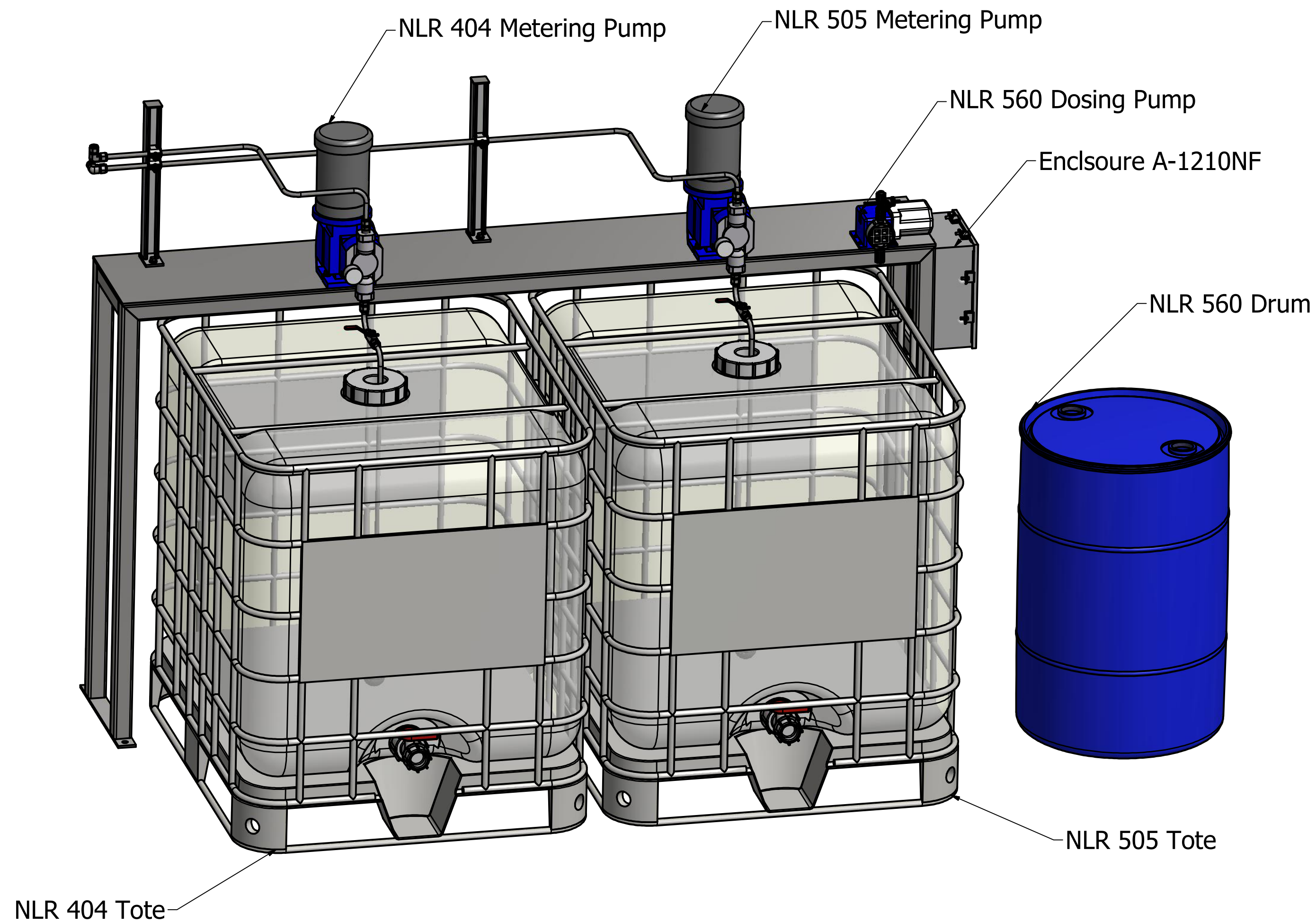
BACK VIEW



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ISO FRONT VIEW



Standard Chemical Metering Station

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4

3

2

1

ISO VIEW

D

D

C

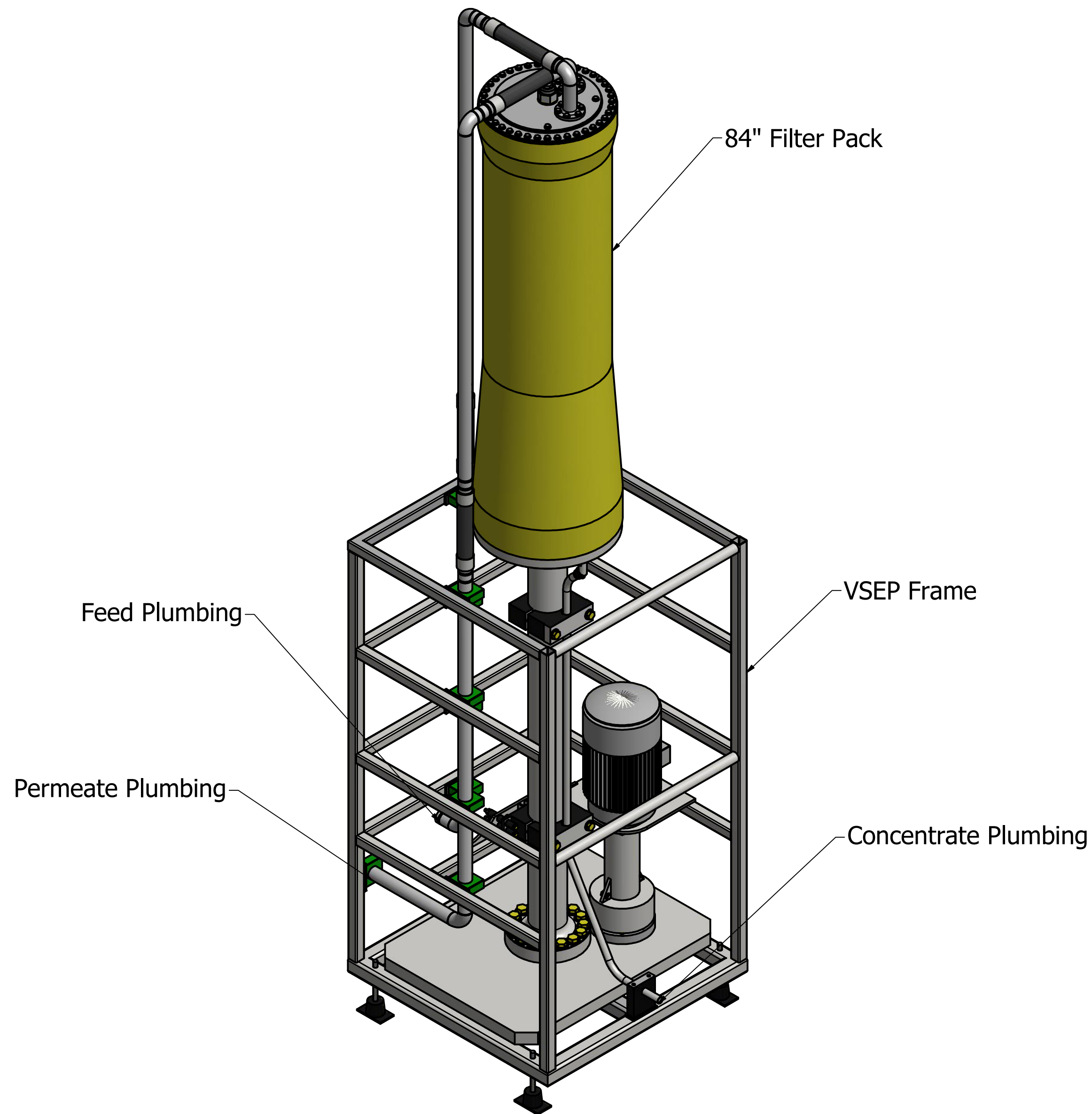
C

B

B

A

A



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Standard 84" VSEP Unit

VSEP SYSTEM COMPONENT SPECIFICATIONS

84" Multiple Module VSEP System
(Applies to 4, 6, 8, and 10 unit systems)

1] Filter Pack:

Membrane:	MF, UF, NF, RO (depends on application and pilot testing)
Allowable Process Temperature:	5°-60°C (some membranes have higher limits up to 120°C)
Max Process Pressure:	550 psi (3,800 kPa)
Allowable Feed Slurry pH Range:	2-12 (some membranes have different limits 1-13)
O-ring and Seal Material:	EPDM (option for Viton)
Filter Pack Housing:	Glass impregnated epoxy resin (FRP)
Wetted Steel Trays:	304 Stainless Steel (option for 316)
Wetted Plastic End Plates:	Polypropylene

2] Vibration System:

Vibration Drive Motor:	Baldor 460VAC, 3phase, 20HP, 3525 RPM
Motor Speed Controller:	Danfoss 20HP 380-480VAC, NEMA 4 (or Equal)
Max. Vibration Amplitude:	3/4 inch (peak to peak measured at the Filter Pack)
Frequency Range:	43.0 Hz to 55.0 Hz (Approximate)
Maximum Decibels:	<85
Bearing RPM:	2800 to 3100
Bearing Oil:	Sullube ~750 ml Compressor Oil
Bearing O-rings:	2-263 & 2-270 Buna

3] Pump Specifications:

Feed Pump Type:	Grundfos 316SS Vertical Multi-stage Centrifugal
Feed Pump Motor:	Grundfos ML 3500 rpm TEFC 460VAC, 3phase
Motor Speed Controller:	Danfoss 380-480VAC, NEMA 4 (or Equal)
Chemical Metering Pump:	Bran+Luebbe Diaphragm pump with manual stroke adjust
Metering Pump Motor:	Baldor 3/4HP TEFC 120/220V, 1phase, 15A, 60/50Hz
Dosing Solenoid Pump:	Chemtech Series 100

4] Pre-Screen Basket Strainer:

Filter Housing Type:	NLR Custom Made 316 SS
Basket Strainer:	100 Mesh 316 SS (Optional Felt Bag Filters)
Capacity:	120 GPM, 4.6 sqft
Elastomers (Seals):	EPDM (option for Viton)

5] Operating Site Conditions:

Equipment Rating:	NEMA 4, Indoor/Outdoor protect from sunlight & rain.
Ambient Temperature:	5° - 37°C
Storage Temperature:	2° - 70°C (Protect Filter Pack from Freezing)
Max Relative Humidity:	<95%, non-condensing
Elevation:	3300 ft maximum without derating VFD/Motor

VSEP SYSTEM COMPONENT SPECIFICATIONS (continued)**6] Piping:**

High Pressure Piping:	Schedule 80 316 SS
Low Pressure Piping:	Schedule 40 316 SS
Special VSEP Concentrate Pipe:	1-1/4" OD 904L Stainless Steel Tubing .065 wall
Flexible Hoses:	SBR/NBR Lined Reinforced Flex Hose
Pipe Clamps:	Zero-Flex Victaulic Style 07 EPDM
Pneumatic Air Supply:	90lb Instrument Grade Air (3/8" connection)
Tubing Fittings:	Hylok 316 SS
On/Off Valves:	316 SS 3-piece ball valve
Manual Valves:	316 SS 3-piece ball valve
Flow Control Valves:	316 SS V-port ball valve
Check Valves:	316 SS swing check valve
Pneumatic Actuators:	Compact 4, Spring Return
Positioners:	BLX Linear Quarter Turn I/P
Skid Pipe Terminations:	150 lb 316 SS Raised Face Flange
Flange Gaskets:	150 lb Garlock Blue Guard

7] CIP:

CIP Pump Type:	Grundfos 316SS Vertical Multi-stage Centrifugal
CIP Pump Motor:	Grundfos ML 3500 rpm TEFC 460VAC, 3phase
Motor Speed Controller:	Danfoss 380-480VAC, NEMA 4 (or Equal)
Clean in Place Tank:	200 Gallon Sloped Cross Linked Polypropylene (36" ø x 54")
pH Meter:	GF Signet 3-2726-00, 3-2750-1, 3-9900-1
Level Sensors:	IFM Efector PN-2298

8] Electrical:

Controls Voltage:	120/220VAC, 1phase, 15A, 50/60Hz
Enclosures:	Hoffman NEMA 4 Powdercoat Finish
Wireway Gutters:	Hoffman NEMA 4 Powdercoat Finish
Switches, Relays, & Contactors:	Allen Bradley 120/220 Volt
Remote Access:	Ewon Cosy 131
Programmable Controller:	AB CompactLogix 1769
Operator Display:	AB Versa View 5400
IO Link Masters and Hubs:	IFM Efector AL1122
Pressure Sensors:	IFM Efector PN-2292
Mag Flow Tube and Transmitter:	IFM Efector SM2000, SM8000
Bag Filter DP Sensors:	IFM Efector PN-2294
Temperature Sensor:	IFM Efector TN-2511, UT0028
Conductivity Meter:	IFM Efector LDL-100, LDL-210
pH Meter:	GF Signet 3-2726-00, 3-2751-1, 3-9900-1
Solenoid Valves:	Festo Manifold Gang Mounted Dual 3/2 24 VDC, IO Link
CIP Communication:	Festo Ethernet Flex IO
VFD Communication:	Ethernet Com Card
DCS Communication:	Ethernet OPC Server (option for Modbus)

Utility Summary VSEP System								New Logic Research	
AIR COMPRESSOR SIZING INFORMATION									
	# of Valves		In^3 Air/stroke		Strokes/day		CFM		M3/hr
VSEP									
1" Pneumatic Valves			16		16.5		20		0.0021 0.0036
1.5" Pneumatic Valves			2		35		20		0.0006 0.001
2" Pneumatic Valves			32		60		20		0.0154 0.0262
2.5" Pneumatic Valves			5		94		20		0.0038 0.0064
3" Pneumatic Valves			2		94		20		0.0015 0.0026
4" Pneumatic Valves			2		112		20		0.0018 0.0031
1" Flow Control Valves			8		35		20		4.0023 6.8038
2" Flow Control Valves			8		60		20		4.0039 6.8066
System Air Totals							System Totals		8.03 13.653
Supply instument grade dry air at 80-90 psi (560-630kPa)									
Supply air to enclosure mounted regulator/filter 3/8" NPT connection on Pump and CIP skids									
Totals do not include air for tools or other sources.									
HOT WATER SIZING INFORMATION									
	# /Day		Temp degC		Gallons/Day		GPM		M3/hr
VSEP									
Cleanings			4		50-60		4000		2.78 0.6306
Rinse filter pack at 60gpm for 5mins			1		50-60		150		0.10 0.0236
Intermittent need of additional cleaning or flush of filter pack			0.14		50-60		21		0.01 0.0033
System Water Totals							System Totals		2.90 0.6575
Supply chlorine free water at >300 uS/cm									
Supply water at 50-60degC and 30gpm (6.8m3/hr)									
Supply water at 15 psi to 2" Hot Water flange connection on CIP skid									
ELECTRICAL SIZING INFORMATION									
Circuit Breaker Sizing 1.25-2x Nameplate Current									
3-Phase Power Voltage VAC			400-460		50/60 Hz				
1-Phase Power Voltage VAC			220-120		50/60 Hz				
	# Motors		Rated Motor HP		Nameplate Current		Nameplate Current		Recommended CB Sizing
					400V		460V		
VSEP Drive Motor			8		20		26.3		24.0 40
VSEP Feed Pump			6		30		38.7		36.0 70
CIP Pump			1		20		26.3		24.0 40
Totals			15						
Supply 3-Phase power from customer supplied load panel (with circuit breakers, disconnects, line filters, etc.) to the inlet of each VFD provided by New Logic. Note two of six VSEP Feed Pumps are hot spares and will be offline until needed.									
	# Connections		Rated Motor		Nameplate		Nameplate		Recommended
					220V		120V		
Control Enclosures			2		NA		10		8.0 20
Metering Pumps			3		0.75		9.2		4.6 15